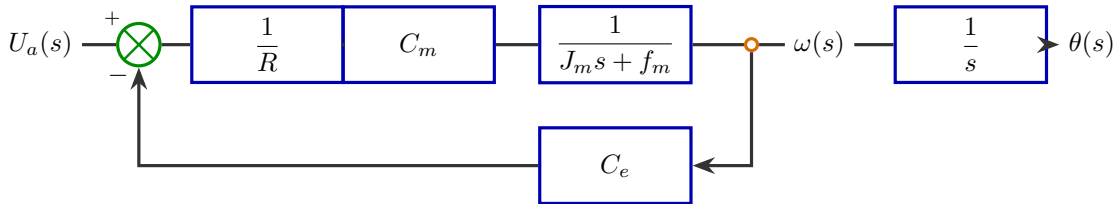
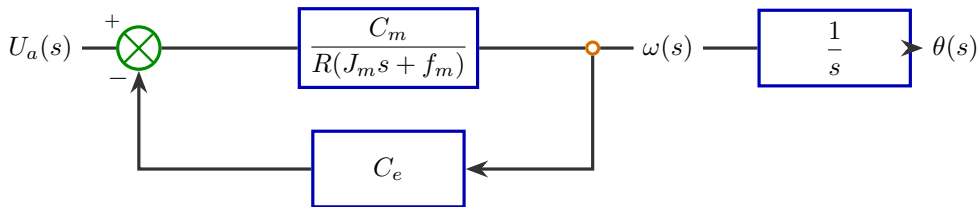


Original DC motor structure



Equivalent reduced form



$$\frac{\omega(s)}{U_a(s)} = \frac{C_m}{RJ_ms + Rf_m + C_eC_m}$$

$$\frac{\theta(s)}{U_a(s)} = \frac{C_m}{s(RJ_ms + Rf_m + C_eC_m)}$$